ADVANCED INDUSTRIAL ORGANIZATION

LECTURE 3

Dynamic Aspects of Imperfect Competition

Sequential decisions, entry, concentration and firm turnover

Study notes based on the lectures of Hui Shen

Xucheng CHEN 陈绪成

The Chinese University of Hong Kong, Shenzhen

October 2026

Reading guide

These notes follow Lecture 3, based on Hui Shen's slides dated October 14, 2026. The section and subsection structure follows the course outline; smaller explanatory headings develop the material within that structure. Progressive builds are consolidated without omitting their additional arguments, examples or calculations.

Original **Model numbers** identify the classroom models; each is followed by a descriptive name. Course figures are redrawn with their axes and economic interpretation. Numbered takeaways separate the conclusions, their mechanisms and the assumptions that support them.

A model is read in the order

$$\text{context} \longrightarrow \text{choices} \longrightarrow \text{optimization} \longrightarrow \text{equilibrium} \longrightarrow \text{meaning}.$$

For a sequential model, start with the last decision and work backwards. For a derivative, identify the variable being changed and what remains fixed. A first-order condition is followed by the relevant feasibility, boundary and deviation checks.

Definitions and results

Blue boxes state the conclusions and the assumptions under which they hold. A displayed optimum is a conclusion about the specified model, rather than a rule for every industry.

Economic mechanism

Green boxes explain the incentives and distinguish a direct effect from an equilibrium response. In particular, a price reduction is a transfer unless it also changes quantities, matching or other real outcomes.

Examples, exercises and lecture cases

Purple boxes contain calculations or applications. Historical cases retain their original period and numerical context. An exercise's hypothetical assumptions are distinguished from historical facts.

Conditions and boundary cases

Amber boxes keep the qualifications that matter for a result: market coverage, nonnegative quantities, integer entry, credible commitment, or an approximation that neglects a firm's effect on an aggregate.

Derivations are beside the model. Boxes labelled "Lecture derivation" develop the classroom calculations and proofs, including those placed in the slides' appendix. Intermediate algebra and explanatory steps are included where needed; no separate mathematical appendix is required to follow the argument.

Instructor stars. The symbol $\star$ preserves an instructor-starred topic. It identifies the course's starred or supplementary material, not an additional importance ranking. Starred sections retain their star in the contents, and the notes state when the slides identify them as skipped.

Notation

Symbol	Meaning and local convention
q_i, $Q = \sum_i q_i$	Firm output and total output.
$R_i(\cdot)$	A follower's best-response function; a function, not just its equilibrium value.
$\pi^g(n)$, $\pi(n)$	Operating profit before entry cost, and net profit after entry cost.
e, f	Entry cost e; f is a subsidy in Section 2.4 but a recurring operating cost in Section 3.2.
n^e, n^*	Free-entry firm count and welfare-maximizing firm count; continuous unless integers are explicitly imposed.
τ	Linear mismatch/transportation cost parameter.
γ, σ	In the CES model: expenditure share and elasticity of substitution, $0 < \gamma < 1 < \sigma$.
$\widetilde{q}$, $\widetilde{p}$	CES consumption aggregate and its exact price index.
M, $R = M\gamma y$	Consumer mass and total spending in the quality-augmented Cournot model.
s_i, $C(s_i) = \alpha s_i^{\beta}$	Quality and its fixed investment cost. Here s_i is quality, not market share.
λ	Common price per quality unit in quality-augmented Cournot.
μ, G, c^*	Active-firm measure, distribution of fresh cost draws, and exit cutoff.
δ, α	In the dynamic model: discount factor and cost persistence. The local use of α differs from the quality-cost parameter.
$V(c)$, $\overline{V}(c)$	Value with the exit option and value conditional on staying.

Three distinctions. A firm can be better off as a leader than under simultaneous play and still prefer to be the follower. Entry cost is paid once before learning or competing; a recurring fixed cost is paid again whenever the firm operates. Stationarity means a constant aggregate distribution, not that every individual firm survives forever.

Contents

Reading guide i

Notation ii

Overview 1

1 Sequential choice: Stackelberg 2

1.1 One leader and one follower . . . 2

1.2 One leader and an endogenous number of followers . . . 8

1.3 Commitment . . . 10

2 Free entry: endogenous number of firms 12

2.1 Properties of free-entry equilibria . . . 12

2.2 Welfare properties of the Cournot model with free entry . . . 14

2.3 Welfare properties of price competition with free entry . . . 17

2.4 Welfare properties of monopolistic competition . . . 20

3 Industry concentration and firm turnover 24

3.1 Exogenous vs. endogenous sunk costs . . . 24

3.2 Dynamic firm entry and exit* . . . 30

Source concordance and references 35

Overview

Lecture 3 · source PDF pp. 1–7

Lecture 2 mostly fixed the number of firms and examined a single competition stage. This lecture adds time in three distinct ways:

1. **Sequential choice: who moves first?** A follower observes an earlier action. Compare leadership and followership, distinguish price from quantity choices, and ask whether the earlier action is credible.
2. **Free entry: how many firms operate?** Entry precedes competition. Compare the private entry decision with the socially desirable count, taking subsequent price or quantity behaviour as given.
3. **Industry structure over time: what changes as a market grows?** Distinguish exogenous setup costs from firms' chosen sunk investments; then allow heterogeneous firms to enter and exit repeatedly.

The learning objectives are to explain timing advantages, evaluate entry and welfare, and connect market size with concentration and turnover. The dynamic-entry topic is instructor-starred and identified as skipped in the overview; it is included here.

SECTION 1

Sequential choice: Stackelberg

Lecture 3 · source PDF pp. 8–80

"Simultaneous" means that a firm cannot condition its choice on an observed rival choice, not necessarily that the two choices occur at precisely the same instant. A sequential game changes this information structure: a follower observes an earlier action before choosing. The earlier action also needs commitment value. If the first mover can freely revise it after the follower acts, it may not influence the follower in the intended way.

The patent-holder/generic-drug example illustrates the timing question: a branded producer may establish its market policy before generic producers enter. Whether the incumbent benefits depends on the strategic variable, on the number of followers, and on whether its policy can actually be maintained.

1.1 One leader and one follower

Lecture 3 · source PDF pp. 9–54

1.1.1 What does a timing advantage compare?

For the same firm with the same technology, compare its payoff when it is the leader with its payoff in the corresponding game when it is the follower. A *first-mover advantage* means that the former is larger; a *second-mover advantage* means that the latter is larger. Comparing the leader with its simultaneous-game payoff is a separate comparison. Both comparisons will be made below.

A strategy for the follower specifies an action *after every possible leader choice*, including choices that do not occur on the equilibrium path. A subgame-perfect equilibrium requires optimal continuation play in all those subgames. Backward induction first derives this entire reaction function and then uses it in the leader's problem.

1.1.2 Quantity competition: the Stackelberg calculation

Model 1: Quantity leadership with one follower

Environment and timing. Firm 1 chooses an irreversible quantity $q_1 \geq 0$. Firm 2 observes it and chooses $q_2 \geq 0$. Demand is $P(Q) = a - Q$ on the positive-price region, with $a > 0$, and both firms have zero marginal costs. The leader anticipates the follower's best response.

Lecture derivation: Backward induction in the quantity game

Stage 2: optimize holding q_1 fixed. The follower solves

$$\max_{q_2 \geq 0} \pi_2 = (a - q_1 - q_2)q_2.$$

Lecture derivation: Backward induction in the quantity game (continued)

In the positive-output region,

$$\frac{\partial \pi_2}{\partial q_2} = a - q_1 - 2q_2 = 0, \qquad \frac{\partial^2 \pi_2}{\partial q_2^2} = -2 < 0.$$

Including the nonnegativity constraint gives the reaction function

$$R_2(q_1) = \max\left\{\frac{a - q_1}{2}, 0\right\}. \tag{1.1}$$

It is a function of the observed q_1, not an independent number chosen by the leader.

Stage 1: substitute the response before differentiating. For $0 \le q_1 \le a$, the leader's profit along the follower's response is

$$\begin{aligned}\pi_1^L(q_1) &= \left[a - q_1 - \frac{a - q_1}{2}\right] q_1 \\ &= \frac{1}{2}(a - q_1)q_1.\end{aligned}$$

Thus

$$\frac{\mathrm{d}\pi_1^L}{\mathrm{d}q_1} = \frac{a}{2} - q_1 = 0 \quad \Longrightarrow \quad q_1^L = \frac{a}{2}.$$

The second derivative is -1. Choosing $q_1 \ge a$ cannot improve on this positive-profit solution under nonnegative market prices.

Recover the follower, price and profits.

$$q_2^F = R_2(a/2) = \frac{a}{4}, \quad Q^S = \frac{3a}{4}, \quad p^S = \frac{a}{4}, \quad \pi_1^L = \frac{a^2}{8}, \quad \pi_2^F = \frac{a^2}{16}. \tag{1.2}$$

For comparison, simultaneous Cournot choices solve $q_1 = (a - q_2)/2$ and $q_2 = (a - q_1)/2$, giving

$$q_i^C = \frac{a}{3}, \qquad p^C = \frac{a}{3}, \qquad \pi_i^C = \frac{a^2}{9}.$$

Lesson 1: quantity leadership in the lecture model

1. **Leadership versus followership:** $\pi^L = a^2/8 > \pi^F = a^2/16$. Each symmetric firm prefers to lead.
2. **Sequential versus simultaneous play:** $q^L = a/2 > q^C = a/3 > q^F = a/4$ and $\pi^L > \pi^C = a^2/9 > \pi^F$. The leader gains and the follower loses relative to Cournot.
3. **Mechanism and scope:** observable output expansion makes the follower contract. The standard extension to nonlinear demand, costs and asymmetric firms requires the corresponding strategic-substitute, concavity and equilibrium conditions.

1.1.3 Why commitment changes the derivative

The difference from Cournot is not an additional physical production advantage. It is that the leader accounts for how the follower changes output. Let $\pi_1(q_1, q_2) = P(q_1, q_2)q_1 - C_1(q_1)$. In a simultaneous game, firm 1 treats q_2 as fixed when varying q_1. In a sequential game, its relevant objective is the composite function $\pi_1(q_1, R_2(q_1))$.

Lecture derivation: The extra strategic term

The total derivative along the reaction function is

$$\frac{\mathrm{d}\pi_1(q_1, R_2(q_1))}{\mathrm{d}q_1} = \underbrace{P + q_1\frac{\partial P}{\partial q_1} - C_1'(q_1)}_{\text{direct effect, holding } q_2 \text{ fixed}} + \underbrace{q_1\frac{\partial P}{\partial q_2}\frac{\mathrm{d}R_2(q_1)}{\mathrm{d}q_1}}_{\text{effect through the follower}} . \tag{1.3}$$

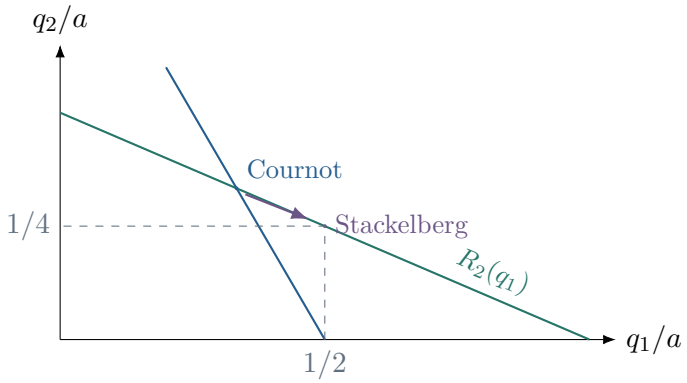
At a Cournot equilibrium (q_1^C, q_2^C), the direct term is zero and $R_2(q_1^C) = q_2^C$. Therefore

$$\left.\frac{\mathrm{d}\pi_1^L}{\mathrm{d}q_1}\right|_{q_1^C} = q_1^C\frac{\partial P}{\partial q_2}\left.\frac{\mathrm{d}R_2(q_1)}{\mathrm{d}q_1}\right|_{q_1=q_1^C} > 0$$

when $\frac{\partial P}{\partial q_2} < 0$ and $\mathrm{d}R_2/\mathrm{d}q_1 < 0$. Increasing own output makes the follower contract, which partly protects the price from the leader's expansion. Under the regularity conditions that give a well-behaved leader optimum, this moves the leader beyond its Cournot quantity.

A larger quantity can make the rival accommodate

The leader gains direct sales but also depresses price. In Cournot those effects balance at $a/3$. With an observable commitment, the follower reduces its quantity when the leader expands. The extra positive strategic term means $a/3$ is no longer the leader's optimum.



The leader selects a point on the follower's reaction curve. Cournot instead requires both firms' fixed-rival best responses to hold at the same time.

1.1.4 Homogeneous-product price competition: a limiting benchmark

Model 2: Sequential homogeneous-product Bertrand pricing

Suppose $c_1 < c_2$, all demand goes to the cheapest seller, and ties go to the more efficient firm. The familiar cost-based comparison selects a low-cost seller's price at c_2 when its unconstrained monopoly price is at least c_2. A leader setting more than the rival's cost invites undercutting; a low-cost leader can retain the market by pricing at that cost. Reversing the order gives the same selected allocation and operating profits. This benchmark therefore need not generate a timing advantage.

Price ceilings, tie rules and continuous undercutting

If the efficient firm's monopoly price p_1^m is below c_2, its selected price is $\min\{c_2, p_1^m\}$, not automatically c_2. Moreover, "set a price just below the rival" describes a supremum and need not be an attained best response with continuous prices. The cost-based comparison is a limiting benchmark; a claim of pure subgame perfection requires a continuation equilibrium in every off-path subgame. A specified price grid or another explicit regularization must be analysed with its own tie and deviation rules.

1.1.5 Differentiated-product price competition

Model 3: Differentiated-product price leadership

Now let symmetric differentiated demands be $Q_i(p_i, p_j)$, with substitutable products and zero marginal costs. Suppose the relevant best responses are smooth and upward-sloping. At simultaneous Bertrand prices, $Q_1 + p_1 \frac{\partial Q_1}{\partial p_1} = 0$. If firm 1 leads, it chooses p_1 anticipating $p_2 = R_2(p_1)$.

Lecture derivation: Leadership raises price when price responses are complementary

$$\frac{\mathrm{d}\pi_1^L}{\mathrm{d}p_1} = \underbrace{Q_1 + p_1 \frac{\partial Q_1}{\partial p_1}}_{\text{direct price effect}} + \underbrace{p_1 \frac{\partial Q_1}{\partial p_2} \frac{\mathrm{d}R_2(p_1)}{\mathrm{d}p_1}}_{\text{rival-price effect}}.$$

At (p_1^B, p_2^B), the direct term is zero. For substitutes, $\frac{\partial Q_1}{\partial p_2} > 0$; strategic complementarity gives $\mathrm{d}R_2/\mathrm{d}p_1 > 0$. Therefore the derivative is positive at the simultaneous equilibrium, and the regular leader optimum has a higher price.

For the lecture's linear demand notation,

$$Q_i = \widetilde{a} - \widetilde{b} p_i + \widetilde{d} p_j, \qquad \widetilde{a} > 0, \quad 0 < \widetilde{d} < \widetilde{b},$$

the follower's problem yields

$$Q_2 - \widetilde{b} p_2 = 0 \quad \Longrightarrow \quad R_2(p_1) = \frac{\widetilde{a} + \widetilde{d} p_1}{2\widetilde{b}}.$$

Its slope is between zero and 1/2. Simultaneous pricing gives

$$p^B = \frac{\widetilde{a}}{2\widetilde{b} - \widetilde{d}}.$$

After substitution, the leader's demand is

$$Q_1(p_1, R_2(p_1)) = \frac{\widetilde{a}(2\widetilde{b} + \widetilde{d})}{2\widetilde{b}} - \frac{2\widetilde{b}^2 - \widetilde{d}^2}{2\widetilde{b}} p_1.$$

Lecture derivation: Leadership raises price when price responses are complementary (continued)

Write this demand as $A - Bp_1$, where

$$A = \frac{\widetilde{a}(2\widetilde{b} + \widetilde{d})}{2\widetilde{b}}, \qquad B = \frac{2\widetilde{b}^2 - \widetilde{d}^2}{2\widetilde{b}} > 0.$$

Then $\pi_1^L = Ap_1 - Bp_1^2$, so

$$\frac{\mathrm{d}\pi_1^L}{\mathrm{d}p_1} = A - 2Bp_1 = 0, \qquad \frac{\mathrm{d}^2\pi_1^L}{\mathrm{d}p_1^2} = -2B < 0.$$

Thus the maximizing price is $A/(2B)$.
Maximizing price times this demand gives

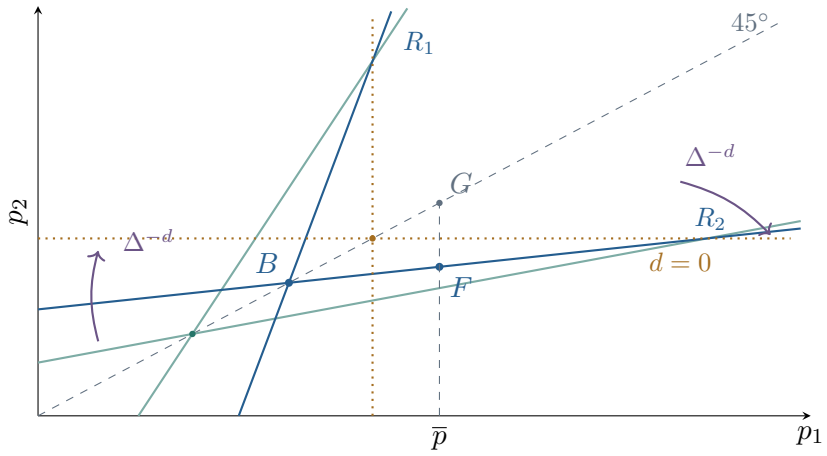
$$p_1^L = \frac{\widetilde{a}(2\widetilde{b} + \widetilde{d})}{2(2\widetilde{b}^2 - \widetilde{d}^2)}, \qquad p_2^F = \frac{\widetilde{a} + \widetilde{d}p_1^L}{2\widetilde{b}}. \tag{1.4}$$

These prices satisfy $p_1^L > p_2^F > p^B$. In particular,

$$p_1^L - p^B = \frac{\widetilde{a}\widetilde{d}^2}{2(2\widetilde{b}^2 - \widetilde{d}^2)(2\widetilde{b} - \widetilde{d})} > 0.$$

The original quadratic-utility coefficients and these direct-demand coefficients are different: $\widetilde{b} = b/(b^2 - d^2)$ and $\widetilde{d} = d/(b^2 - d^2)$. The condition $b > d > 0$ implies the inequalities used above.

Reading the course diagram. In the (p_1, p_2) plane, firm 2's response is flatter than the 45° line. Firm 1's response, written as a line in the same plane, is steeper. Their intersection B lies on the diagonal by symmetry. At a leader price $\overline{p} > p^B$, the follower chooses a point $F = (\overline{p}, R_2(\overline{p}))$ below the diagonal, so $R_2(\overline{p}) < \overline{p}$.



Course figure: price competition with substitutes. Source PDF p. 50. The source's additional lines are interior reaction equations at different product-interaction parameters. Δ^{-d} means a decrease in d. The underlying utility coefficients are fixed at $a = 10/3, b = 25/21$; green, blue and amber use $d/b = 0.7, 0.4, 0$. Blue therefore gives direct demand $Q_i = 2 - p_i + 0.4p_j$. At the fixed blue parameter, B is Bertrand and $F = (\overline{p}, R_2(\overline{p}))$ is below the diagonal. The added point $G = (\overline{p}, \overline{p})$ explains the subsequent payoff comparison.

The arrows Δ^{-d} in the source describe a *change in the product-interaction parameter*. They rotate or shift both response curves and change their intersection. This differs from the leadership comparison: to compare B, F and G, hold the demand parameters fixed and use the blue response. A change in the rival's price at fixed parameters raises own demand, which justifies the profit comparison below.

1.1.6 Why the follower can earn more than the leader

Let the leader set $\overline{p} > p^B$, and write $R(\overline{p}) < \overline{p}$, as in the linear example. By symmetry, use one profit function $\pi(p_i, p_j)$. The follower optimizes against $\overline{p}$, so

$$\pi(R(\overline{p}), \overline{p}) > \pi(\overline{p}, \overline{p}).$$

For a positive-sales firm, a higher rival price raises profit. Since $\overline{p} > R(\overline{p})$,

$$\pi(\overline{p}, \overline{p}) > \pi(\overline{p}, R(\overline{p})).$$

Combining them gives

$$\underbrace{\pi(R(\overline{p}), \overline{p})}_{\text{follower}} > \underbrace{\pi(\overline{p}, R(\overline{p}))}_{\text{leader}}.$$

The intermediate expression is a comparison payoff, not a third equilibrium. Both firms may gain relative to simultaneous Bertrand while the follower gains more.

An explicit check of the two comparisons

With $Q_i = 2 - p_i + 0.4p_j$ and zero costs,

$$p^B = 1.25, \quad p_1^L \simeq 1.30435, \quad p_2^F \simeq 1.26087,$$

$$\pi^B = 1.56250, \quad \pi_1^L \simeq 1.56522, \quad \pi_2^F \simeq 1.58979.$$

Leadership improves the leader's payoff relative to simultaneous play, but being the follower is even more attractive. This illustrates why the reference payoff must be stated.

Lesson 2: price leadership and second-mover advantage

1. **Symmetric differentiated firms:** $\pi^F > \pi^L$. The follower retains flexibility and prices below the leader; both can nevertheless gain relative to simultaneous Bertrand.
2. **Cost asymmetry:** sufficiently similar unit costs preserve second-mover advantage for both firms. With sufficiently different costs in the lecture's maintained price-game class, the low-cost firm can prefer leadership while the high-cost firm prefers followership.
3. **General lesson:** at least one firm has a second-mover advantage under the lecture's demand/cost assumptions. The result concerns the actual signs of demand and response effects; complementary goods require those signs to be reconsidered.

The direction of the strategic response matters, rather than the name "price" or "quantity" alone. For complementary products the cross effects can reverse; the reasoning must be reapplied to their actual signs.

1.2 One leader and an endogenous number of followers

Lecture 3 · source PDF pp. 55–60, 302–309

A leader facing free entry anticipates an additional response: its action changes how many firms enter. The patent-expiration example now includes multiple potential generic producers, each evaluating whether its post-entry profit covers a setup cost. This differs from the earlier single-follower pricing game. An aggressive incumbent policy can reduce entry as well as change each follower's action.

There are three distinct points in the course's argument:

1. **Motivation:** after patent expiry, the former patent holder may cut price while generic producers consider entry. This is modelled as imperfect competition with strategic entrants, rather than a passive competitive fringe.
2. **Two response channels:** the incumbent affects both each follower's market action and the number of followers. The second channel is absent from Model 3.
3. **Qualified prediction:** under suitable demand, cost and entry conditions, leadership becomes more aggressive: greater output in quantity models or a lower price in price models. The following CES calculation establishes the price case explicitly; it does not assume the statement for every demand system.

The following price model makes that mechanism explicit. It uses the lecture's appendix calculation here, immediately after the motivating discussion. Followers' individual pricing derivatives hold their number fixed; the leader's derivative allows the number to adjust. These are two different optimization exercises.

Environment and timing. The incumbent commits to p_L. Potential followers pay entry cost $f > 0$ and then set prices simultaneously, observing p_L. All firms have constant marginal cost $c > 0$. Consumer spending on differentiated goods is E, CES elasticity is $\sigma > 1$, and the number n of followers is approximated as continuous. The incumbent's own setup cost is already sunk.

Lecture derivation: CES price leadership with free entry

Demand and the unilateral follower derivative. Define

$$q_i = \frac{Ep_i^{-\sigma}}{D}, \qquad D = p_L^{1-\sigma} + \sum_{j=1}^{n} p_j^{1-\sigma}, \qquad s_i = \frac{p_i q_i}{E} = \frac{p_i^{1-\sigma}}{D}.$$

Here s_i is an expenditure share. A follower's price change affects D even though n is held fixed:

$$\frac{\partial \log D}{\partial \log p_i} = (1-\sigma)s_i, \qquad \eta_i = -\frac{\partial \log q_i}{\partial \log p_i} = \sigma - (\sigma-1)s_i.$$

The FOC for $(p_i - c)q_i - f$ is therefore

$$\frac{p_i - c}{p_i} = \frac{1}{\eta_i}. \tag{1.5}$$

It would be incorrect to impose elasticity σ on these non-negligible followers at a fixed number of firms.

At a fixed number of firms, raising rivals' prices increases firm i's expenditure share, reduces η_i, and raises its desired markup. This is the appendix's strategic-complementarity observation. It concerns the unilateral pricing problem before free entry is imposed.

Lecture derivation: CES price leadership with free entry (continued)

Use zero profit to determine follower price and share. In a symmetric follower equilibrium,

$$f = (p_F - c)q_F = \frac{Es_F}{\sigma - (\sigma - 1)s_F}.$$

Rearranging gives

$$\sigma f = [E + (\sigma - 1)f]s_F,$$

so

$$s_F = \frac{\sigma f}{E + (\sigma - 1)f}, \qquad p_F = \frac{\sigma cE}{(\sigma - 1)(E - f)}. \tag{1.6}$$

Assume $E > f$. These values depend on primitives, not on the committed leader price. Free entry absorbs that change.

Derive the entry response. Since $s_F = p_F^{1-\sigma}/D$, free entry fixes $D = \overline{D} = p_F^{1-\sigma}/s_F$. Let $N = 1/s_F$. Dividing the expression for D by $p_F^{1-\sigma}$ gives

$$N = \left(\frac{p_F}{p_L}\right)^{\sigma-1} + n, \quad \Longrightarrow \quad n(p_L) = N - \left(\frac{p_F}{p_L}\right)^{\sigma-1}.$$

Therefore

$$n'(p_L) = (\sigma - 1)\frac{p_F^{\sigma-1}}{p_L^{\sigma}} > 0. \tag{1.7}$$

A lower incumbent price leaves room for fewer followers.

Optimize along the entry response. Because $\overline{D}$ remains fixed along this response,

$$\pi_L^g(p_L) = \frac{E}{\overline{D}}(p_L - c)p_L^{-\sigma},$$

$$\frac{\mathrm{d}\pi_L^g}{\mathrm{d}p_L} = \frac{E}{\overline{D}}p_L^{-\sigma-1}\left[\sigma c - (\sigma - 1)p_L\right].$$

The sign is positive before the root and negative after it, so

$$p_L^* = \frac{\sigma c}{\sigma - 1} < p_F = p_L^*\frac{E}{E - f}, \qquad \frac{q_L}{q_F} = \left(\frac{p_F}{p_L^*}\right)^{\sigma} > 1. \tag{1.8}$$

Without the leader there are N symmetric free-entry firms. With it,

$$1 + n(p_L^*) = N + 1 - \left(\frac{E}{E - f}\right)^{\sigma-1} < N.$$

The incumbent's larger share displaces more than one symmetric follower's share in the continuous approximation.

The lecture's numerical illustration

Let $\sigma = 2$, $c = 1$, $E = 9$ and $f = 1$. Then

$$s_F = \frac{1}{5}, \quad N = 5, \quad p_F = \frac{9}{4}, \quad p_L^* = 2, \quad s_L = \frac{9}{40}, \quad n^* = \frac{31}{8}.$$

The quantities and operating profits are

$$q_F = \frac{4}{5}, \quad q_L = \frac{81}{80}, \qquad \pi_F^g = 1 = f, \quad \pi_L^g = \frac{81}{80}.$$

If the leader is charged the same setup cost, its net profit is $1/80$. The total count falls from 5 to $39/8$; fractional counts reflect the approximation, not literal fractional firms.

Positive-entry region and global scope

A sufficient condition for the displayed leader optimum to remain inside a positive-entry continuation for every $p_L \geq c$ is

$$N > 1 + \left(\frac{p_F}{c}\right)^{\sigma-1}.$$

It implies $n(c) > 1$; $p_L < c$ gives negative operating profit. In the numerical illustration this is $5 > 13/4$. If entry can be completely deterred, compare that regime separately. Integer entry uses entry/no-entry inequalities, not exact zero profit. The lower-leader-price result is established here for this CES environment; it is not a universal consequence of any form of endogenous entry.

1.3 Commitment

Lecture 3 · source PDF pp. 61–80

A threat or promise changes a rival's action only if it is credible. Schelling (1960) explains why an announcement may be cheap talk: when execution becomes necessary, the player might prefer to reverse it. A useful commitment has three requirements:

1. **Visible:** the rival observes it. Perfect information supplies this in the two-stage model.
2. **Understandable:** the rival correctly interprets the action and its implications. Rational interpretation is maintained.
3. **Credible:** carrying out the action remains optimal or reversal is unavailable/costly. The two-stage timing assumes that the leader cannot freely undo its action in a later third stage.

Lecture derivation: A third-stage revision destroys the original quantity commitment

At the two-stage Stackelberg outcome, $q_1 = a/2$ and $q_2 = a/4$. Suppose firm 1 can revise its quantity after observing q_2. Its new problem, holding $q_2 = a/4$ fixed, gives

$$q_1 = \frac{a - q_2}{2} = \frac{3a}{8} < \frac{a}{2}.$$

The originally announced $a/2$ is not sequentially optimal in the added stage.
If the initial announcement is entirely costless and has no further effect, firm 2 anticipates $q_1 = (a - q_2)/2$ at the final stage. It then maximizes $\frac{1}{2}(a - q_2)q_2$, choosing $q_2 = a/2$;

Lecture derivation: A third-stage revision destroys the original quantity commitment (continued)

firm 1 finally chooses $a/4$. The last irreversible move, rather than the first announcement, determines leadership in this altered game.

The paradox of commitment

Restricting one's future options can improve the current outcome by changing someone else's best response. "Burning the bridge" is the lecture's military analogy: eliminating retreat can make a determination to fight credible. The economic requirement is the changed future incentive, not the announcement itself.

The lecture offers three economic sources of commitment:

1. **Capacity and technology.** A plant designed for a particular use is hard to redeploy. Interpreting quantity leadership through capacity requires an appropriate connection between installed capacity and later output; the two are not automatically identical.
2. **Price-adjustment restrictions.** A regulated firm may need approval to change its price while its unregulated rival can respond freely. A printed catalogue can be costly to replace, unlike an Internet price list. Regulation and a prior choice of sales technology can therefore change timing.
3. **Contracts.** A most-favoured-customer clause extends favourable terms to other covered buyers. Cutting a price for one buyer can trigger rebates to others, changing the incentive to cut later. In the lecture's suitable contractual setting, the proposer becomes the pricing leader in a Stackelberg-like outcome; the precise effect depends on the clause and trading environment.

1.3.1 Strategic choices and tactical choices

An earlier *strategic* choice changes the environment of later *tactical* choices such as price and quantity. Capacity investment, selling technology and contract design are examples. With a fixed rival count, quantity substitutes favour an aggressive commitment to expand, while complementary price responses can make a soft, high-price commitment attractive to a leader even though following is preferable. Endogenous entry adds the further incentive to deter or limit entry. Always identify which continuation game the strategic choice shapes.

SECTION 2

Free entry: endogenous number of firms

Lecture 3 · source PDF pp. 81–182

Previously the number of competitors was fixed, implicitly because additional entry was prohibitively costly or excluded from the short-run analysis. Free entry instead means that potential firms may enter by paying the stated entry cost; it does not mean that entry is costless. The game has an entry stage followed by price or quantity competition. Solving the second stage determines the profit opportunity that drives the first.

An entry cost paid before competition is sunk in that competition stage. It still matters for the earlier entry decision. Distinguish operating profit $\pi^g(n)$, before the setup payment, from net profit $\pi(n) = \pi^g(n) - e$. The lecture sometimes uses one symbol for both in different models; the distinction is explicit here.

2.1 Properties of free-entry equilibria

Lecture 3 · source PDF pp. 82–102, 310–313

Model 4: Symmetric free entry followed by competition

2.1.1 Entry and no-entry conditions

Suppose the second-stage symmetric operating profit decreases with the number of firms, $\pi^g(n+1) < \pi^g(n)$. With an integer firm count, a free-entry equilibrium n^e satisfies

$$\pi^g(n^e) \geq e, \qquad \pi^g(n^e + 1) \leq e. \tag{2.1}$$

The first inequality says that an active entrant covers its cost. The second says that another firm would not gain by joining, after allowing competition to adjust. Equalities create indifference and potentially adjacent admissible counts. If even a monopolist cannot cover entry cost, $n^e = 0$.

With continuous entry and a positive interior count, the two conditions are approximated by $\pi^g(n^e) = e$. Higher entry costs then reduce the equilibrium count. Entry normally takes time, making this a long-run comparison; low-cost virtual entry can sometimes justify a shorter-run interpretation.

Case 1: entry in small US cities

1. **Question and setting.** Bresnahan and Reiss (1990, 1991) study how homogeneous-product firm counts respond to demand in separate rural retail and professional markets. The lecture reports an average town population of about 3,740.
2. **Method.** An ordered-probit entry model treats firms within each market as identical competitors. Limited scope for spatial differentiation helps motivate that approximation in these selected towns.
3. **Finding and mechanism.** Entry requires a margin sufficient to cover operating fixed costs; additional entry drives margins down. Entry thresholds combine demand, costs and conduct, so counts alone do not identify each component separately.

2.1.2 Why twice the market need not support twice the firms

A proportional increase in market size multiplies demand at *every given price*. For $Q = (a-p)/b$, doubling market size means replacing b with $b/2$ in inverse demand, not doubling the intercept a. Write $d = 1/b$ as the market-size parameter for this calculation:

$$P(Q) = a - \frac{Q}{d}.$$

Larger d makes the price reduction caused by an extra unit of output smaller. Firms expand, competition lowers margins, and firms need more sales to cover the same entry cost.

Lecture derivation: Market size and firm count in linear Cournot

For n symmetric firms with constant marginal cost $c < a$, the quantity-game FOC is

$$a - c - \frac{(n+1)q}{d} = 0 \quad \Longrightarrow \quad q(n) = \frac{d(a-c)}{n+1}.$$

The margin is $(a-c)/(n+1)$, so operating profit is

$$\pi^g(n) = \frac{d(a-c)^2}{(n+1)^2} = \frac{q(n)^2}{d}.$$

Continuous zero profit gives

$$n^e = (a-c)\sqrt{\frac{d}{e}} - 1, \qquad q(n^e) = \sqrt{de}. \tag{2.2}$$

Differentiate the count, treating a, c, e as fixed:

$$\frac{\partial n^e}{\partial d} = \frac{a-c}{2\sqrt{de}} > 0, \qquad \frac{\partial n^e}{\partial d}\frac{d}{n^e} = \frac{n^e+1}{2n^e}.$$

The elasticity is at most one for $n^e \geq 1$, and strictly below one for $n^e > 1$. At large counts it approaches $1/2$. A continuous solution below one is not a literal positive integer industry and must be checked against the monopoly entry condition.

Lesson 3: market expansion and competitive effects

1. **More firms:** increasing market size raises the viable count in the maintained homogeneous-good entry environment.
2. **More competition:** expansion changes firms' incentives and lowers the equilibrium margin. Holding price fixed would omit this effect.
3. **Larger firms:** to cover unchanged entry cost at a lower margin, each firm needs more sales. In linear Cournot, $q^e = \sqrt{de}$ and the count elasticity is $(n^e + 1)/(2n^e)$, below one for $n^e > 1$.

2.1.3 Why consumer gains do not settle the entry-welfare question

More firms can lower prices and, with differentiated products, improve variety. But additional firms also duplicate setup costs and may increase average production cost by operating at smaller scale. A welfare comparison therefore asks what additional *real surplus* an entrant creates relative to the extra resources it uses. Revenue captured from existing firms is not all new social value. The next three subsections apply that distinction to Cournot, spatial price competition and monopolistic competition.

2.2 Welfare properties of the Cournot model with free entry

Lecture 3 · source PDF pp. 103–127, 314

Model 5: Cournot free entry and second-best welfare

Environment and timing. Firms first pay a fixed setup cost $e > 0$ and then compete in quantities. A second-best planner can choose the number of entrants but cannot change their subsequent Cournot behaviour. Let $q(n)$ be each firm's symmetric equilibrium output, $Q(n) = nq(n)$ total output, $P(Q)$ inverse demand and $C(q)$ variable production cost.

The argument uses three maintained assumptions:

A1. $q'(n) < 0$: each incumbent contracts when another firm enters (business stealing).

A2. $Q'(n) > 0$: total industry output nevertheless increases.

A3. $P(Q(n)) > C'(q(n))$: selling an additional unit has positive social value at the equilibrium margin.

Also take downward-sloping demand $P' < 0$ and the differentiability needed below. The proof will use each assumption separately.

Net profit is

$$\pi(n) = P(nq(n))q(n) - C(q(n)) - e.$$

Ignoring the integer constraint, free entry solves $\pi(n^e) = 0$. The planner's objective is

$$W(n) = \int_0^{nq(n)} P(s)\,\mathrm{d}s - nC(q(n)) - ne. \tag{2.3}$$

The integral is consumers' gross willingness to pay. Subtract variable resource costs and setup costs. Price payments cancel between consumers and firms, so they are not subtracted a second time.

Lecture derivation: Business stealing and the welfare derivative

For an integral with a moving upper limit, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}n}\int_0^{Q(n)} P(s)\,\mathrm{d}s = P(Q(n))Q'(n), \qquad Q'(n) = q(n) + nq'(n).$$

This is the relevant case of Leibniz's formula. More generally,

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} h(x,t)\,\mathrm{d}t = h(x,v(x))v'(x) - h(x,u(x))u'(x) + \int_{u(x)}^{v(x)} \frac{\partial h(x,t)}{\partial x}\,\mathrm{d}t.$$

Continuity of the integrand and its derivative and differentiable limits justify this expression. Differentiate each remaining term in (2.3):

$$\begin{aligned} W'(n) &= P(Q)\{q + nq'\} - C(q) - nC'(q)q' - e \\ &= \underbrace{P(Q)q - C(q) - e}_{\pi(n)} + n\{P(Q) - C'(q)\}q'(n). \end{aligned}$$

Hence

$$\boxed{W'(n) = \pi(n) + n\big[P(nq(n)) - C'(q(n))\big]q'(n).} \tag{2.4}$$

This decomposition has two parts:

1. **Direct contribution:** the entrant contributes its own net profit $\pi(n)$.
2. **Effect on incumbents:** their total contraction is $nq'(n)$, valued at the social output margin $P - C'$. The product $n(P - C')q'$ is the external effect missing from the entrant's calculation.

Because the margin is positive and $q'(n) < 0$, the second term is negative. The marginal entrant values entry more highly than the planner does:

$$W'(n) < \pi(n).$$

The negative term refers to the output contraction of the n pre-existing firms; it does not say total industry output falls. Total output increases by $q + nq' > 0$ under the maintained assumption.

To complete the count comparison, differentiate net profit:

$$\pi'(n) = \{P(Q) - C'(q)\}q'(n) + q(n)P'(Q)Q'(n) < 0.$$

At a regular interior planner optimum, $W'(n^*) = 0$, so $\pi(n^*) > 0$. Since profits decrease to zero at n^e, it follows that $n^e > n^*$.

Lesson 4: excessive entry under the maintained Cournot assumptions

1. **Private versus social value:** business stealing gives $W'(n) = \pi(n) + n(P - C')q'(n) < \pi(n)$.
2. **Continuous-count conclusion:** if an interior planner optimum exists and net profit decreases with entry, free entry produces $n^e > n^*$.
3. **Integer qualification:** actual counts must satisfy entry/no-entry inequalities and welfare finite differences. Under-entry can occur in special integer cases; the continuous derivative cannot be used as a rounding rule.

2.2.1 Linear demand: all quantities and counts explicitly

Let $P(Q) = a - bQ$, $C(q) = cq$ and $a > c$, $b > 0$. The symmetric Cournot solution is

$$q(n) = \frac{a-c}{b(n+1)}, \qquad Q(n) = \frac{n(a-c)}{b(n+1)}, \qquad p(n) - c = \frac{a-c}{n+1}.$$

Consequently $q' < 0$, $Q' = (a-c)/[b(n+1)^2] > 0$ and the margin is positive for every finite n. Net profit and welfare are

$$\pi(n) = \frac{(a-c)^2}{b(n+1)^2} - e, \tag{2.5}$$

$$W(n) = \frac{(a-c)^2}{2b}\left[1 - \frac{1}{(n+1)^2}\right] - ne. \tag{2.6}$$

For the welfare expression, insert $Q(n)$ into $(a-c)Q - bQ^2/2 - ne$ and simplify. Thus

$$\pi(n^e) = 0 \implies (n^e+1)^2 = \frac{(a-c)^2}{be},$$

$$W'(n^*) = \frac{(a-c)^2}{b(n^*+1)^3} - e = 0 \implies (n^*+1)^3 = \frac{(a-c)^2}{be}.$$

Writing $A = (a-c)^2/(be)$, the positive interior formulas are

$$n^e = \sqrt{A} - 1, \qquad n^* = A^{1/3} - 1. \tag{2.7}$$

For $A > 1$, $\sqrt{A} > A^{1/3}$. If parameters put an optimum at a boundary, compare the boundary rather than retaining a negative count.

The lecture's three-versus-seven illustration

Take $A = 64$. Then the continuous planner count is $n^* = 3$ and the free-entry count is $n^e = 7$. Direct comparison of integer welfare also selects three in this example. It demonstrates economically meaningful duplication of entry costs even when the count must be an integer.

2.2.2 Integer entry: use finite differences

Continuous zero profit cannot be used to round a planner optimum without checking actual welfare. For integer $n \geq 1$,

$$W(n) - W(n-1) = e\left[\frac{A(2n+1)}{2n^2(n+1)^2} - 1\right]. \tag{2.8}$$

This is the welfare gain from moving from $n-1$ to n Cournot firms, not the derivative at a fractional count. The gains decrease with n, so choose the largest n for which the gain remains nonnegative, allowing ties.

For instance, a monopolist enters privately only when $A \geq 4$, while a planner prefers one firm to none when $A \geq 8/3$. Hence for $8/3 < A < 4$, the privately chosen count is zero while the socially preferred count is one. The firm cannot appropriate all the consumer surplus it creates.

Parameter region (away from ties)	Free entry	Planner
$A < 8/3$	0	0
$8/3 < A < 4$	0	1
$4 < A < 9$	1	1
$9 < A < 72/5$	2	1
$72/5 < A < 16$	2	2
$16 < A < 25$	3	2

At thresholds, specify how indifferent entrants are treated. The general Cournot business-stealing result with integers allows entry to be below the planner count by at most one under its maintained assumptions; the linear example's under-entry region is the zero-versus-one case. The exact inequalities and finite differences, rather than a single cutoff for all larger A, determine equality or excessive entry.

Entry taxes or permits can in principle address excessive entry. Broadcasting and mobile-telephony permits also reflect a distinct reason for restriction: spectrum scarcity. The welfare calculation above is an economic argument under a model, not a general recommendation to restrict entry in every market.

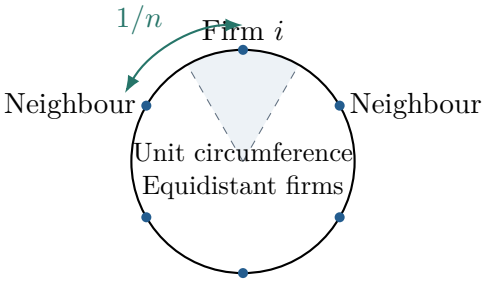
2.3 Welfare properties of price competition with free entry

Lecture 3 · source PDF pp. 128–147

Model 6: Salop price competition with free entry

Environment and timing. Consumers of unit mass are uniformly distributed on a circle of circumference one. They buy one unit, have sufficiently high reservation value, and incur linear transport cost τ per unit distance. Entrants pay setup cost e and are required to locate equidistantly. They then choose prices. Marginal cost is c. Location choice itself is not optimized in this entry calculation.

A circle can represent shops along a ring road or airline departure times on a 24-hour clock. The distance between neighbours is $1/n$. In the usual symmetric local-competition pricing region, each firm serves $1/n$ and sets $p(n) = c + \tau/n$.



Annotated course sketch: the Salop circle. Source PDF p. 129 and the discussion on pp. 130–138. Six firms illustrate equidistant entry; n is general in the algebra. Each firm serves an arc of length $1/n$, extending $1/(2n)$ in each direction at symmetric prices.

Lecture derivation: Price equilibrium, free entry and the planner

Recover the pricing condition. If the neighbouring prices equal p, a small unilateral price change by firm i moves each of its two boundaries by $-(p_i - p)/(2\tau)$. Adding both sides gives

$$Q_i = \frac{1}{n} + \frac{p - p_i}{\tau}, \qquad \frac{\partial Q_i}{\partial p_i} = -\frac{1}{\tau}.$$

The FOC is $Q_i - (p_i - c)/\tau = 0$, yielding at symmetry

$$p(n) - c = \frac{\tau}{n}, \qquad \pi(n) = \frac{\tau}{n^2} - e.$$

Under the maintained symmetric price-equilibrium and coverage conditions, continuous free entry therefore gives

$$n^e = \sqrt{\frac{\tau}{e}}, \qquad p^e = c + \sqrt{\tau e}. \tag{2.9}$$

Measure the planner's real costs. With everyone buying one unit, production cost and gross product valuation are constant. Prices only transfer income. Each firm serves a segment of length $1/n$, extending $1/(2n)$ on either side. Total transport cost is

$$2n \int_0^{1/(2n)} \tau x \, \mathrm{d}x = 2n \left[\frac{\tau}{2} x^2 \right]_0^{1/(2n)} = \frac{\tau}{4n}.$$

The planner minimizes setup plus transport cost:

$$TC(n) = ne + \frac{\tau}{4n}.$$

Since $TC'(n) = e - \tau/(4n^2)$ and $TC''(n) = \tau/(2n^3) > 0$,

$$n^* = \frac{1}{2}\sqrt{\frac{\tau}{e}} = \frac{1}{2} n^e. \tag{2.10}$$

Lesson 5: variety benefits do not rule out excessive entry

1. **Private entry:** $n^e = \sqrt{\tau/e}$ in the maintained covered, equidistant, continuous Salop calculation.
2. **Efficient entry:** $n^* = \frac{1}{2}\sqrt{\tau/e}$ minimizes setup cost plus transport cost, so $n^e = 2n^*$.
3. **Reason:** better matching is a real benefit, but capturing existing customers is not all new surplus. Price reductions alone are transfers because aggregate purchases remain fixed at one.

Higher e gives fewer firms and a higher equilibrium price. Higher τ both raises margins and supports more entrants: products are harder to substitute, so a firm can cover setup cost with fewer customers. The continuous formulas are meaningful in their multi-firm region; integer counts and cases with very few firms require the corresponding price subgame.

Why a lower price has a different welfare role here

In Cournot, a lower price accompanies increased total output and can reduce underconsumption. In the covered Salop model, total quantity remains one. A lower price benefits consumers but reduces firm revenue by the same amount. The real entry benefits are shorter travel and better matching, while setup costs are real losses. Consumer-surplus improvement and total-surplus improvement are different statements.

Case 2: US radio stations

1. **Market and data.** Berry and Waldfogel (1999) study 135 US metropolitan radio markets. Station revenue is annual advertising revenue per listener times listeners; larger populations can support more stations and programming variety.
2. **Demand and revenue.** They estimate listening participation and inverse advertising demand from advertising prices, station counts and listening data. These determine how station revenue changes with entry.
3. **Costs and welfare.** Free entry is used to infer the distribution of fixed station costs. Revenues and costs give the modelled surplus of stations and advertisers; the second-best planner changes station numbers, not subsequent conduct.
4. **Result.** The lecture reports excessive entry and a welfare loss around 40% of industry revenue relative to that optimum. This is the specified model's historical calculation, not an automatic valuation of all listener benefits or a universal percentage loss.

2.4 Welfare properties of monopolistic competition

Lecture 3 · source PDF pp. 148–182, 315–316

2.4.1 Why this is a different differentiated-product model

Chamberlin's (1933) *monopolistic competition* combines four features:

1. **Many varieties:** a large number of firms each produces one differentiated variety.
2. **Negligible individual firms:** a firm affects others through aggregate demand rather than direct strategic interaction with a particular rival.
3. **Free entry and exit:** no additional barriers prevent entry; positive entry costs are covered and net economic profit is zero.
4. **Individual market power:** each variety faces downward-sloping demand, so price can exceed marginal cost.

The first three features resemble competition; the fourth resembles monopoly. Restaurants, clothing, shoes and urban services motivate this hybrid. Zero net profit means that operating profit pays for setup cost, not that the markup vanishes.

The Spence–Dixit–Stiglitz model (Spence, 1976; Dixit and Stiglitz, 1977) differs from Salop in two ways. Each variety competes through a common aggregate with all other varieties, rather than just two neighbours. A representative consumer buys some of every variety because of love of variety, rather than heterogeneous consumers each choosing one product. Entry does not alter the CES substitution parameter between a given pair of varieties.

Model 7: Spence–Dixit–Stiglitz monopolistic competition

Environment and timing. A consumer has income y and utility $U = q_0^{1-\gamma}\widetilde{q}^{\gamma}$, $0 < \gamma < 1$. The numeraire has price one. The differentiated composite is $\widetilde{q} = (\sum_{i=1}^{n} q_i^{(\sigma-1)/\sigma})^{\sigma/(\sigma-1)}$, $\sigma > 1$. Firms pay setup cost e and have constant marginal cost $c > 0$. In pricing, each small firm treats the aggregate price index and expenditure as fixed.

Lecture derivation: Two consumer problems and the CES price index

Allocate income between the numeraire and the composite. For composite price $\widetilde{p}$, the budget binds at $q_0 = y - \widetilde{p}\widetilde{q}$. Maximizing log utility is equivalent to maximizing utility:

$$\max_{\widetilde{q}} \; (1-\gamma)\log(y - \widetilde{p}\widetilde{q}) + \gamma \log \widetilde{q}.$$

The FOC is

$$-\frac{(1-\gamma)\widetilde{p}}{y - \widetilde{p}\widetilde{q}} + \frac{\gamma}{\widetilde{q}} = 0 \quad \Longrightarrow \quad \widetilde{p}\widetilde{q} = \gamma y, \quad q_0 = (1-\gamma)y.$$

Thus γy is spending on the entire differentiated sector, not spending on each variety.

Allocate sector expenditure across varieties. The expenditure allocation has its own Lagrangian,

$$\mathcal{L} = \widetilde{q} + \zeta \left[\gamma y - \sum_j p_j q_j \right].$$

Its partial derivative is $\frac{\partial \mathcal{L}}{\partial q_i} = \frac{\partial \widetilde{q}}{\partial q_i} - \zeta p_i$. Setting it to zero equates marginal utility to the multiplier times price; it does not make a raw utility derivative equal to zero.

Lecture derivation: Two consumer problems and the CES price index (continued)

Maximize $\widetilde{q}$ subject to $\sum_i p_i q_i = \gamma y$. With multiplier ζ, the FOC is

$$\frac{\partial \widetilde{q}}{\partial q_i} = \widetilde{q}^{1/\sigma} q_i^{-1/\sigma} = \zeta p_i, \quad \Longrightarrow \quad q_i = \widetilde{q}\zeta^{-\sigma} p_i^{-\sigma}.$$

Let $D = \sum_j p_j^{1-\sigma}$. Substituting into the expenditure constraint yields

$$\widetilde{q}\zeta^{-\sigma} D = \gamma y.$$

Substituting the same FOCs into the definition of the composite gives

$$\widetilde{q} = \left[\sum_i (\widetilde{q}\zeta^{-\sigma} p_i^{-\sigma})^{(\sigma-1)/\sigma}\right]^{\sigma/(\sigma-1)} = \widetilde{q}\zeta^{-\sigma} D^{\sigma/(\sigma-1)}.$$

After cancelling $\widetilde{q} > 0$, $\zeta^{-\sigma} = D^{-\sigma/(\sigma-1)}$. Together with expenditure this identifies the exact index:

$$\widetilde{p} = D^{1/(1-\sigma)} = \left(\sum_j p_j^{1-\sigma}\right)^{1/(1-\sigma)}. \tag{2.11}$$

Demand for each variety is therefore

$$q_i = \frac{\gamma y p_i^{-\sigma}}{\sum_j p_j^{1-\sigma}} = \widetilde{p}^{\sigma-1}\gamma y\, p_i^{-\sigma}. \tag{2.12}$$

At equal prices, $\widetilde{p} = p n^{-1/(\sigma-1)}$. Adding varieties lowers the cost of a unit of the composite even with the individual price held fixed.

2.4.2 From consumer demand to pricing and entry

The ratio $q_i/q_j = (p_i/p_j)^{-\sigma}$ makes σ the elasticity of substitution. For a *negligible* firm holding $\widetilde{p}$ fixed, its own-price demand elasticity is also σ. Consequently the profit FOC is

$$q_i + (p_i - c)\frac{\partial q_i}{\partial p_i} = 0 \quad \Longrightarrow \quad \frac{p_i - c}{p_i} = \frac{1}{\sigma}.$$

Therefore

$$p = \frac{\sigma}{\sigma - 1}c, \qquad q(n) = \frac{(\sigma-1)\gamma y}{\sigma n c}, \qquad \pi(n) = \frac{\gamma y}{\sigma n} - e. \tag{2.13}$$

Continuous free entry gives

$$n^e = \frac{\gamma y}{\sigma e}, \qquad q(n^e) = \frac{(\sigma-1)e}{c}. \tag{2.14}$$

The price is independent of sector spending in this specification. More spending supports more varieties; higher setup cost or greater substitutability reduces the count. Each firm still charges a positive markup, with its operating profit exactly funding setup cost.

The small-firm approximation is a maintained assumption

Equation (2.12) contains all prices through the index. A non-negligible firm differentiating that finite-variety demand exactly has elasticity $\sigma - (\sigma - 1)s_i$, as in Section 1.2. The monopolistic-competition calculation uses the explicit negligible-firm approximation. The two elasticity formulas answer different pricing problems and should not be interchanged.

2.4.3 The second-best planner and a variety subsidy

The planner can affect entry but retains the firms' markup pricing. A subsidy f per firm is financed by a lump-sum tax nf on the consumer, giving disposable income $y - nf$. A negative f is an entry tax. Keep the financing in the consumer budget; a subsidy is not a free resource.

Lecture derivation: Eliminate the policy variable, then choose entry

At symmetric prices, sector spending is $\gamma(y - nf)$, so

$$q = \frac{(\sigma-1)\gamma(y-nf)}{\sigma n c}, \qquad \widetilde{q} = qn^{\sigma/(\sigma-1)} = \frac{(\sigma-1)\gamma(y-nf)}{\sigma c}\, n^{1/(\sigma-1)}.$$

Since $q_0 = (1-\gamma)(y-nf)$,

$$U = Kn^{\gamma/(\sigma-1)}(y-nf), \qquad K = (1-\gamma)^{1-\gamma}\left[\frac{(\sigma-1)\gamma}{\sigma c}\right]^{\gamma} > 0.$$

For given policy, free entry requires

$$\frac{\gamma(y-nf)}{\sigma n} - e + f = 0.$$

Multiply by σn and solve for the policy that implements a chosen n:

$$f = \frac{n\sigma e - \gamma y}{n(\sigma-\gamma)}, \qquad y - nf = \frac{\sigma}{\sigma-\gamma}(y - ne).$$

Substitution gives the planner's objective

$$U(n) = \frac{K\sigma}{\sigma-\gamma} n^{\gamma/(\sigma-1)}(y-ne), \qquad y - ne > 0.$$

Writing $k = \gamma/(\sigma-1) > 0$, its derivative has the sign of

$$k(y-ne) - ne.$$

It is zero at

$$n^* = \frac{\gamma y}{(\sigma - 1 + \gamma)e} > \frac{\gamma y}{\sigma e} = n^e, \tag{2.15}$$

because $0 < \gamma < 1$. The derivative changes from positive to negative there. The implementing subsidy is

$$f^* = \frac{(1-\gamma)e}{\sigma-\gamma} > 0.$$

Lesson 6: entry can be insufficient as well as excessive

1. **Result in this specification:** $n^* = \gamma y/[(\sigma - 1 + \gamma)e] > n^e = \gamma y/(\sigma e)$. The planner wants more varieties.
2. **Mechanism:** a new variety creates benefits that its entrant cannot fully appropriate. The implementing subsidy is positive and must be financed in the consumer budget.
3. **Broader lesson:** another monopolistic-competition specification can give excessive entry. What matters is appropriated new surplus relative to business stealing and real setup costs.

The contrast with Salop does not arise from an algebraic inconsistency: consumers, substitution patterns and the effect of variety on utility have changed. The welfare conclusion belongs to the entire model, not to the label "differentiated products" alone.

Entry model	What entry changes	Continuous-count result
Cournot	Total output rises; incumbents contract	Excessive entry under A1–A3
Covered Salop	Matching improves; total output is fixed	$n^e = 2n^*$
Specified CES model	Extra varieties enter the consumption composite	$n^e < n^*$

How to use the comparison. First identify what an entrant creates; next identify which part it can charge for; finally deduct real production and setup costs. A larger consumer surplus by itself does not settle total welfare.

SECTION 3

Industry concentration and firm turnover

Lecture 3 · source PDF pp. 183–300

The entry models so far ask whether an equilibrium count is too high or too low. This section asks a different question: how does market size change industry structure? First, setup costs are taken as exogenous. Then firms choose quality investments that are sunk before market competition. Finally, the instructor-starred dynamic model allows ongoing entry and exit with heterogeneous changing costs.

3.1 Exogenous vs. endogenous sunk costs

Lecture 3 · source PDF pp. 184–247

3.1.1 Exogenous setup costs and fragmentation

A setup cost e is exogenous if the firm's decisions do not change its size, and sunk if it cannot be recovered upon exit. With demand multiplied by market size M, a larger market can support more firms for a given setup cost. In the linear Cournot illustration, replace b with b/M:

$$P(Q) = a - \frac{b}{M}Q, \qquad n^e + 1 = (a-c)\sqrt{\frac{M}{be}}.$$

For a fixed market, reducing e towards zero likewise makes the Cournot count unbounded. The distinction is between a *given* setup payment and a sunk investment chosen in response to market opportunities.

As $M \to \infty$, $n^e \to \infty$. At symmetry each share is $1/n$, normalized Herfindahl concentration is $H = 1/n$, and the top-k share is k/n when $n \geq k$. These measures tend to zero.

Lesson 7: exogenous sunk costs in the lecture's comparable-quality models

1. **Expansion supports entry:** in the lecture's comparable-quality models, fixed exogenous setup cost does not escalate with market size.
2. **Concentration can vanish:** unbounded entry gives $H = 1/n \to 0$ and top-k share $k/n \to 0$.
3. **Empirical interpretation:** Sutton studies a lower envelope across comparable markets, not a universal exact curve. Exogenous sunk cost alone is not enough to exclude every natural oligopoly; Lecture 4 gives a demand-based exception.

Across real markets, many demand, technology and institutional conditions vary. Sutton's lower-bound approach consequently studies the least concentrated outcomes attainable at comparable market sizes, rather than fitting one exact firm-count curve to every observation. In exogenous-sunk-cost industries that lower envelope approaches zero. Hair salons are an example discussed in the lecture.

Persistently concentrated industries, both over periods of demand growth and across differently sized local markets, motivate another mechanism. A larger market may induce firms to spend more on a competitive advantage, raising an *endogenous* sunk investment rather than only supporting more firms at unchanged setup cost.

3.1.2 Quality-augmented Cournot: what consumers and firms choose

Model 8: Quality investment, quantity competition and natural oligopoly

Environment and timing. There are M consumers, each with income y and Cobb–Douglas preferences between the numeraire and quality-adjusted consumption. A share γ of income is spent on this industry, so its total revenue is $R = M\gamma y$. Firms first enter at cost e, then choose qualities $s_i > 0$ at fixed investment cost $C(s_i)$, and finally choose physical quantities q_i. Marginal physical production cost is $c > 0$.

The Cobb–Douglas spending claim follows from a consumer problem. At a given price λ of quality-adjusted consumption S, the budget is $q_0 + \lambda S = y$. After substitution, maximize $(y - \lambda S)^{1-\gamma} S^{\gamma}$. Its interior FOC, or the derivative of its logarithm, gives

$$-\frac{(1-\gamma)\lambda}{y - \lambda S} + \frac{\gamma}{S} = 0 \quad \Longrightarrow \quad \lambda S = \gamma y.$$

Multiplying expenditure per consumer by the consumer mass M yields $R = M\gamma y$. At the market level, total quality-adjusted output is the sum supplied by all firms.

Consumers treat one quality-adjusted unit as interchangeable across suppliers. A convenient aggregate is $S = \sum_i s_i q_i$, with utility proportional to $q_0^{1-\gamma} S^{\gamma}$. If active products had different quality-per-dollar ratios s_i/p_i, consumers would purchase only from products offering the better ratio. Positive sales by several firms therefore require

$$\frac{p_i}{s_i} = \lambda \quad \text{for every active firm,} \qquad R = \sum_i p_i q_i = \lambda \sum_i s_i q_i. \tag{3.1}$$

Thus inverse demand is the system

$$\lambda = \frac{R}{\sum_j s_j q_j}, \qquad p_i = \lambda s_i. \tag{3.2}$$

Prices are market-clearing consequences of quantities and qualities. Firms do not choose prices again in this quantity stage. The quality-adjusted homogeneity assumption also differs from the heterogeneous-consumer vertical model in Lecture 4.

Why fixed industry spending does not fix an individual firm's revenue

Cobb–Douglas preferences fix the total R spent on this sector. A firm can still win a larger portion of that spending. Raising its quantity changes λ, hence all quality-adjusted prices. Raising quality before competition changes the firm's attractiveness per unit produced. Aggregate expenditure stays fixed while revenue shares change.

3.1.3 Stage 3: solve quantities conditional on all qualities

What is held fixed? The quality vector $(s_1, \ldots, s_n)$, revenue R and the number of active firms are fixed. The common price-quality ratio λ is not fixed when a firm varies q_i: equation (3.2) makes it endogenous. Entry and quality costs are already sunk.

Lecture derivation: Eliminate the common price-quality ratio

Let $S = \sum_j s_j q_j$. Then

$$\frac{\partial \lambda}{\partial q_i} = -\frac{R s_i}{S^2} = -\frac{s_i \lambda^2}{R}.$$

The firm's operating profit is $(\lambda s_i - c)q_i$. Its FOC is

$$\lambda s_i - c + s_i q_i \frac{\partial \lambda}{\partial q_i} = 0,$$

so

$$\lambda s_i - c = \frac{s_i^2 q_i \lambda^2}{R} \quad \Longrightarrow \quad s_i q_i = \frac{R}{\lambda} - \frac{cR}{\lambda^2 s_i}.$$

Define $A_s = \sum_j 1/s_j$. Sum these n equations and use $\sum_i s_i q_i = R/\lambda$:

$$\frac{nR}{\lambda} - \frac{cR}{\lambda^2} A_s = \frac{R}{\lambda} \quad \Longrightarrow \quad \lambda = \frac{cA_s}{n-1}, \qquad n > 1.$$

Substitution back into the firm-specific equation yields

$$q_i = \frac{R}{c} \frac{n-1}{s_i A_s} \left[1 - \frac{n-1}{s_i A_s} \right]. \tag{3.3}$$

The price-cost margin and operating profit are

$$p_i - c = c \left[\frac{s_i A_s}{n-1} - 1 \right], \tag{3.4}$$

$$\pi_i^g = (p_i - c) q_i = R \left[1 - \frac{n-1}{s_i A_s} \right]^2. \tag{3.5}$$

These are the all-active formulas. Each firm's positive-sales condition is

$$s_i A_s > n - 1.$$

The lowest-quality firm is the most restrictive one. If a calculated q_i is nonpositive, remove the inactive firm and solve the quantity stage again with the active set. Squaring a negative bracket in (3.5) does not make an inactive firm's infeasible allocation valid.

At equal quality s, $A_s = n/s$, so

$$q_i = \frac{R(n-1)}{cn^2}, \qquad p_i = \frac{cn}{n-1}, \qquad \pi_i^g = \frac{R}{n^2}. \tag{3.6}$$

Physical output and price are independent of the common quality level. This follows from fixed total spending and quality-adjusted substitution: a common quality improvement changes utility, but not relative positions or sector spending.

If quality were fixed exogenously at s, entry would satisfy

$$\frac{R}{n^2} - e - C(s) = 0 \quad \Longrightarrow \quad n^e = \sqrt{\frac{M\gamma y}{e + C(s)}}.$$

Then the count grows without bound with M. A fixed quality cost by itself behaves like another exogenous setup payment.

3.1.4 Stage 2: quality investment responds to market size

Now allow each firm to choose its quality before the quantity stage. Hold rivals' common quality $\widehat{s}$ fixed during a unilateral deviation. With $C(s_i) = \alpha s_i^{\beta}$, $\alpha, \beta > 0$, equation (3.5) becomes

$$\pi_i^g - C(s_i) = R\left[1 - \frac{1}{1/(n-1) + s_i/\widehat{s}}\right]^2 - \alpha s_i^{\beta}$$

in the positive-sales region. The firm's own s_i changes its subsequent equilibrium quantity and price, which are already incorporated in this reduced profit.

Lecture derivation: The symmetric quality first-order condition

Write $h(s_i) = 1 - 1/[1/(n-1) + s_i/\widehat{s}]$. Then

$$h'(s_i) = \frac{1/\widehat{s}}{[1/(n-1) + s_i/\widehat{s}]^2}.$$

Differentiate *before* imposing symmetry:

$$2Rh(s_i)h'(s_i) - \alpha\beta s_i^{\beta-1} = 0.$$

At a symmetric interior candidate $s_i = \widehat{s} = s^*$,

$$h(s^*) = \frac{1}{n}, \qquad h'(s^*) = \frac{(n-1)^2}{n^2 s^*}.$$

Therefore

$$\frac{2R(n-1)^2}{n^3 s^*} = \alpha\beta(s^*)^{\beta-1} \quad \Longrightarrow$$

$$s^*(n, R) = \left[\frac{2R(n-1)^2}{\alpha\beta n^3}\right]^{1/\beta}. \tag{3.7}$$

For a fixed n, quality rises with $R = M\gamma y$. The induced investment is

$$C(s^*) = \frac{2R(n-1)^2}{\beta n^3}. \tag{3.8}$$

The entry cost e does not enter this FOC because it is sunk before quality is chosen.

A simultaneous increase in every firm's quality leaves their relative ranking unchanged, yet any one firm would gain a competitive advantage by improving quality while the others did not. This explains an investment escalation without an increase in total sector revenue.

3.1.5 Stage 1: a bound on entry that market size cannot remove

Subtract entry and quality costs from symmetric operating profit:

$$\pi(n) = \frac{R}{n^3}\left[n - \frac{2}{\beta}(n-1)^2\right] - e. \tag{3.9}$$

Positive net profit with $e > 0$ requires the bracket to be strictly positive. Solving its upper-root condition gives

$$n < n_+(\beta) \equiv 1 + \frac{\beta + \sqrt{\beta^2 + 8\beta}}{4}. \tag{3.10}$$

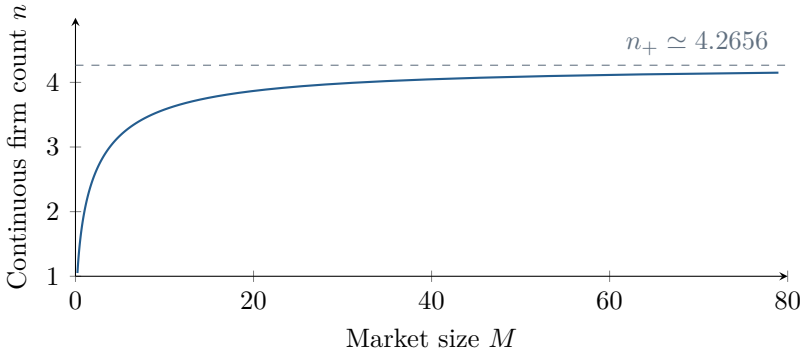
Revenue R multiplies the bracket but does not change its sign. A larger market therefore cannot sustain a symmetric positive-profit candidate with an arbitrarily large count through this mechanism. The slides also give $\beta = 3$ as a simple check: $n - \frac{2}{3}(n-1)^2$ is positive for integer $n = 1, 2, 3$ and negative for $n \geq 4$. The quantity-stage calculation itself requires $n > 1$. Thus at most three active firms pass the positive-profit bound in that parameter illustration.

The lecture's natural-oligopoly illustration

For $\beta = 5$, $n_+ = (9 + \sqrt{65})/4 \simeq 4.2656$, so at most four firms can satisfy the positive-profit condition with integer entry. Using $e = 0.2$ and $\gamma y = 1$, equation (3.9) becomes

$$\pi(n) = \frac{M}{n^3}\left[n - \frac{2}{5}(n-1)^2\right] - 0.2.$$

The continuous zero-profit branch rises with market size towards 4.2656; the integer admissible count cannot exceed four. Market growth raises competitive quality spending as well as the number of firms.



Course figure: firm count and market size. Source PDF pp. 230–232. The rising zero-profit branch is bounded as the market expands; the integer count must be checked separately.

The curve solves the continuous zero-profit equation for the displayed parameters. The dashed line is the upper root of the positive-profit bracket, not another zero-profit firm count at finite market size.

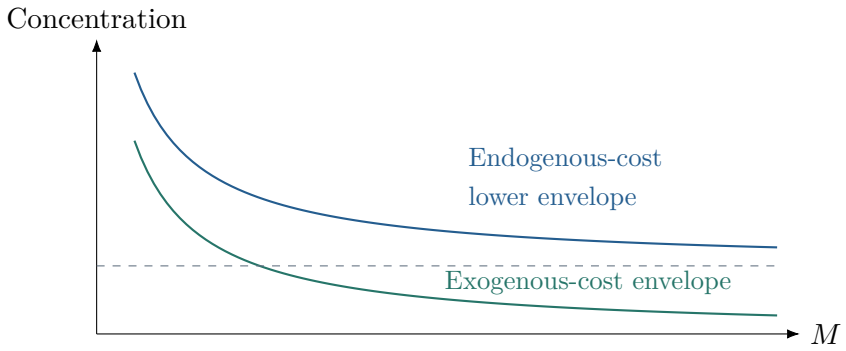
A necessary bound is not a complete equilibrium-existence proof

The calculation derives a symmetric interior quality candidate and a necessary bound for profitable activity. Global quality deviations, the active-set conditions, the existence of the quality subgame, and integer entry/no-entry incentives must also hold for a full equilibrium. In particular the displayed $n > 1$ formulas do not solve a separate monopoly problem. The lecture's concentration mechanism is the bounded viable count, not a claim that every root of a first-order or zero-profit equation is automatically an equilibrium.

Lesson 8: endogenous sunk investment can prevent fragmentation

1. **A larger prize induces investment:** increasing M raises $R = M\gamma y$ and strengthens quality-investment incentives.
2. **Investment absorbs the opportunity:** all firms invest more; relative positions and total sector spending need not improve proportionately. The sunk cost required to compete rises endogenously.
3. **Concentration has a floor:** under the maintained mechanism, the viable firm count is bounded independently of market size. Advertising, product innovation and cost-reducing process innovation can play analogous roles, subject to their own demand and technology conditions.

Sutton's interpretation is a *lower envelope*, not one exact functional relationship for all markets: firms in a given large market can be more concentrated for other reasons, while investment escalation prevents concentration from becoming arbitrarily small. With a bound of $\bar{n}$ active firms, $H \geq 1/\bar{n}$ by the share-squares inequality; symmetry gives $H = 1/n$.



Course figure: concentration lower envelopes. Source PDF pp. 240–241. The exogenous-cost envelope approaches zero; the endogenous-cost envelope approaches a strictly positive floor. The axes are market size and concentration, not a fitted regression line.

The figure is schematic: it contrasts limiting possibilities rather than reproducing an estimated cross-market regression. Lecture 4 also gives a different natural-oligopoly mechanism based on vertical demand characteristics, without requiring investment escalation.

Case 3: US supermarkets and distribution systems

1. **Question and market definition.** Ellickson (2007) studies 1998 US supermarkets, defined in the slides as full-line food stores with annual sales above $2 million. Perishables limit the relevance of nationwide scale. Similar distribution-centre locations help identify 51 distribution markets; about 82% of store revenue comes from products supplied within its assigned market.
2. **Quality and investment.** Quality is assortment. Chains can (i) build larger stores and shelf space or (ii) fit more products into given space through better replenishment and distribution. Logistics software, optical scanners and sorting equipment support the second route; distribution investment became important from the 1980s.
3. **Evidence.** Average product counts rose from roughly 14,000 in 1980 to 22,000 in 1994 and 30,000 in 2004. Four to six chains dominate each distribution market despite differences in market size; some small stores survive.
4. **Interpretation.** Investment costs are mostly local to each distribution market, so nationwide scale alone does not account for concentration. Larger markets support escalated assortment and distribution investment, consistent with endogenous sunk costs. These are the lecture's historical figures.

3.2 Dynamic firm entry and exit*

Lecture 3 · source PDF pp. 248–300

Instructor-starred topic; identified as skipped in the lecture overview. The material is retained in its course position. A static entry model cannot explain why firms enter and exit in the same industry at the same time. Heterogeneous firms with changing prospects can do so: some obtain good cost realizations while others become uncompetitive. Stationarity fixes the aggregate distribution, not the identities of firms.

3.2.1 Timing, states and the active-firm distribution

Model 9: Stochastic entry, exit and a stationary industry

There is a continuum of small firms and a market of size M. A firm's state is its current marginal cost $c \in [0, 1]$ and whether it was active in the previous period. A new entrant draws c from distribution G. An incumbent retains its cost with probability α; with probability $1 - \alpha$ it draws again from G. Assume $0 \leq \alpha < 1$ in the positive-turnover comparison. Firms discount by $0 < \delta < 1$.

Each period has four stages:

1. Potential entrants decide whether to pay sunk entry cost e.
2. New and incumbent firms learn their current costs; an entrant has paid before seeing its draw.
3. Firms choose whether to exit, receiving zero if inactive.
4. Those that remain pay recurring fixed cost f and compete.

An incumbent continuing to the next period does not repay e. It does repay f whenever it operates. This difference is central to the option value of remaining active.

Let μ be the measure of active firms across cost types; $M\pi(c;\mu)$ is operating profit before recurring fixed cost. This is again an **exogenous-sunk-cost industry**: firms are small, there is no positive concentration floor, and the dynamic model is separate from Model 8's investment escalation.

The competition stage is summarized by six assumptions. Write $\mu \succeq \mu'$ when the competitive environment μ is at least as favourable for every relevant type as μ'.

1. **Efficiency ranking:** lower c gives strictly higher operating profit wherever profits are positive.
2. **More competitors worsen the environment:** if μ' contains more firms at every cost level, $\pi(c;\mu) \geq \pi(c;\mu')$ for all relevant c.
3. **A common ordering:** competitive environments are completely ordered by this preference relation. The ranking cannot favour low-cost types in one direction and high-cost types in the other.
4. **Continuity:** the profit function varies continuously in its maintained arguments.
5. **Absolute gains favour efficient firms:** when μ is preferable to μ', $\pi(c;\mu)-\pi(c;\mu')$ strictly decreases with c on the relevant support.
6. **Proportional gains favour inefficient firms:** $\pi(c;\mu)/\pi(c;\mu')$ strictly increases with c where the denominator is positive.

Assumptions 5 and 6 are compatible. For example, a profit increase from 10 to 14 is larger in dollars but smaller proportionally than an increase from 1 to 2. Reversing such a relaxation makes inefficient firms suffer the larger percentage loss.

The lecture links these restrictions to suitable output and price responses and cites continuum versions of Cournot and differentiated linear-demand price competition as examples. These links rely on the specified competition games; they are not generic equivalences for every profit function.

3.2.2 Value of staying and the exit cutoff

Lecture derivation: Bellman equation and the continuation option

At the exit stage, define $V(c)$ as value including the choice to leave, and $\overline{V}(c)$ as value if the firm stays now:

$$V(c) = \max\{0, \overline{V}(c)\}, \tag{3.11}$$

$$\overline{V}(c) = M\pi(c;\mu) - f + \delta\left[\alpha V(c) + (1-\alpha)\int_0^1 V(z)\,\mathrm{d}G(z)\right]. \tag{3.12}$$

The terms are current profit, discounted continuation with the same cost, and discounted continuation after a fresh cost draw. Let c^* be the cutoff: types below it stay, types above it exit, and at an interior continuous cutoff $V(c^*) = \overline{V}(c^*) = 0$.
Positive entry requires an entrant's expected value net of setup cost to be zero:

$$V^e = \int_0^1 V(c)\,\mathrm{d}G(c) - e = 0. \tag{3.13}$$

The expectation includes bad draws with $V(c) = 0$; entrants pay for unsuccessful attempts as well as successful ones.

Lecture derivation: Bellman equation and the continuation option (continued)

For an active type, $V(c) = \overline{V}(c)$ and free entry gives $\int V \, \mathrm{d}G = e$. Substituting in (3.12) and collecting the same-cost continuation term gives

$$(1 - \delta\alpha)\overline{V}(c) = M\pi(c;\mu) - f + \delta(1-\alpha)e,$$
$$\overline{V}(c) = \frac{M\pi(c;\mu) - f + \delta(1-\alpha)e}{1 - \delta\alpha}, \qquad c < c^*.$$

Since $1 - \delta\alpha > 0$, the cutoff satisfies

$$M\pi(c^*;\mu) - f + \delta(1-\alpha)e = 0. \tag{3.14}$$

Free entry can then be written

$$\int_0^{c^*} \left[M\pi(c;\mu) - f + \delta(1-\alpha)e\right] \mathrm{d}G(c) = (1 - \delta\alpha)e. \tag{3.15}$$

For a current type above the cutoff, $V(c) = 0$, so the hypothetical value of staying for this period is the numerator of the active-value expression, without the factor $1/(1 - \delta\alpha)$.

A firm may rationally operate at a current loss

At the cutoff, current profit is

$$M\pi(c^*;\mu) - f = -\delta(1-\alpha)e < 0$$

when $e > 0$, $\delta > 0$ and fresh draws occur. By staying, the firm preserves the possibility of a favourable future cost draw without paying setup cost again. Current profit and the value of the firm are different objects. A firm exits when the loss outweighs this continuation option.

3.2.3 The exit-policy calculation in the slides

For a proposed initial cutoff x, followed by optimal future behaviour, define

$$\mathcal{V}(x;\mu) = \int_0^x \overline{V}(c) \, \mathrm{d}G(c) - e.$$

At equilibrium, $\mathcal{V}(c^*;\mu) = 0$. If G has positive density g at the cutoff,

$$\frac{\partial \mathcal{V}(x;\mu)}{\partial x} = \overline{V}(x)g(x),$$

so its interior FOC at c^* is $\overline{V}(c^*) = 0$. This is the same marginal-type indifference condition as (3.14), expressed through the initial exit-policy problem. It is not an additional independent equation that replaces population stationarity.

3.2.4 Stationarity and two definitions of turnover

Lecture derivation: Balance the flows of cost types

Let $N = \mu([0, c^*])$ be the mass of operating firms and let m be the mass of new entry attempts before their cost draws. A stationary active type measure satisfies, on $[0, c^*]$,

$$\mu(\mathrm{d}c) = \alpha\mu(\mathrm{d}c) + [(1-\alpha)N + m]G(\mathrm{d}c).$$

The first term retains incumbent costs; the second contains incumbent redraws and new attempts. Rearranging and integrating over active types, with $F^* = G(c^*)$, gives

$$(1-\alpha)N = [(1-\alpha)N + m]F^*,$$

$$\frac{m}{N} = (1-\alpha)\frac{1-F^*}{F^*}, \qquad \mu([0,c]) = N\frac{G(c)}{G(c^*)} \quad (0 \le c \le c^*). \tag{3.16}$$

Thus the active distribution is a truncated and rescaled G.
The lecture's turnover ratio is entry attempts per operating firm, m/N. Only the fraction F^* of attempts becomes operating entrants. Actual operating entry and exit per incumbent are therefore

$$h = \frac{mF^*}{N} = (1-\alpha)(1-F^*). \tag{3.17}$$

Both ratios increase as the cutoff falls, but their levels differ. The attempts ratio can exceed one; the incumbent exit probability h cannot.

Together, cutoff indifference, free entry and population balance characterize the stationary distribution. Under the lecture model's additional ordering and regularity assumptions, the positive-turnover stationary equilibrium is unique. This is an attributed equilibrium result; the algebra above derives the value and flow restrictions, rather than proving uniqueness for every possible profit function.

3.2.5 Larger markets, selection and firm age

Holding competitors fixed, more consumers raise every active firm's profit and entry value. Entry then expands the active population, putting downward pressure on prices. Under the maintained comparative-profit assumptions, this competitive effect is proportionally harsher for inefficient firms. The marginal surviving firm must therefore become more efficient:

$$M' > M \quad \Longrightarrow \quad N(M') > N(M), \qquad c^*(M') < c^*(M).$$

A lower cutoff decreases $G(c^*)$, raising both turnover measures in (3.16). The resulting stationary industry has more firms but stronger selection, greater replacement and a younger age distribution. This is a comparison between stationary equilibria, not only a temporary wave of exit after integration. The market-size argument has three steps:

1. **Expansion at fixed competition:** a larger M raises every active firm's revenue opportunity and the value of entry.
2. **Entry and selection:** the active mass rises, intensifying competition. The proportional-profit restriction makes this pressure more severe for inefficient firms, reducing c^*.
3. **Replacement and age:** a lower $G(c^*)$ raises exit risk after a redraw and raises turnover. To see the age implication, with constant stationary exit hazard h, cohort survival through k redraw periods is proportional to $(1-h)^k$. Larger h makes old firms less common.

Lesson 9: market size changes selection as well as firm counts⋆

1. **More active firms:** a larger market supports a larger operating mass.
2. **Stronger selection:** the upper active-cost cutoff falls; only more efficient types survive.
3. **Higher turnover and younger firms:** replacement rises in the new stationary equilibrium, not merely during a transition. The conclusion uses the competition-stage assumptions listed above.

Market integration can therefore eliminate inefficient firms that survived in separate small markets. New entrants have a lower probability of a sufficiently good draw, while successful active firms serve higher volumes. More recurring fixed cost f also requires stronger efficiency and raises turnover under the model's conditions. More sunk entry cost e raises the value of avoiding re-entry and reduces turnover. The distinction between the two costs prevents an apparent contradiction.

Case 4: entry and exit of Swedish hair salons⋆

1. **Why salons?** Asplund and Nocke (2006) study Swedish hair salons: service is local and physical, turnover is frequent, there are many small firms even in small markets, and few chains. These features motivate independent local markets and monopolistic competition without multi-outlet complications.
2. **Measurement.** Postal areas define markets; population proxies market size, with commuting as a possible problem. Land values proxy premises rents, allowing fixed operating costs to differ across markets.
3. **Prediction and test.** A random sample of 1,030 salons is split into small and large markets. The predicted age CDF in larger markets lies above the one in smaller markets: at each age, a greater share of firms is younger.
4. **Evidence.** The lecture reports that the CDF ordering is confirmed and that larger market size has a highly statistically significant negative association with firm age, robust to controls for market heterogeneity. The result supports the model's prediction; this account does not establish a new causal estimate.

Source concordance and references

The following map uses physical pages in the supplied PDF. Progressive builds are consolidated in the indicated discussion; the original appendix calculations appear in the main text beside their models.

PDF pages	Destination	Material
1–7	Overview	Overview and course structure
8–14	§1.1	Timing and information
15–34	§1.1	Quantity leadership and chain rule
35–54	§1.1	Price leadership and timing comparisons
55–60	§1.2	Endogenous followers
61–80	§1.3	Commitment and tactical choices
81–90	§2.1	Entry and no-entry conditions
91–93	§2.1	Case 1: US small cities
94–102	§2.1	Market size and entry welfare motivation
103–117	§2.2	Cournot welfare and business stealing
118–127	§2.2	Linear illustration and integer entry
128–143	§2.3	Salop entry and planner
144–147	§2.3	Case 2: radio broadcasting
148–172	§2.4	Monopolistic competition and CES derivation
173–182	§2.4	Planner financing and variety subsidy
183–199	§3.1	Exogenous sunk cost and concentration
200–220	§3.1	Quality-augmented quantity competition
221–232	§3.1	Endogenous quality, entry and concentration bound
233–242	§3.1	Lower envelopes and investment interpretation
243–247	§3.1	Case 3: supermarkets
248–261	§3.2	Starred dynamic model environment and timing
262–270	§3.2	Competition-stage assumptions
271–281	§3.2	Bellman, free entry and exit-policy derivations
282–294	§3.2	Stationarity, turnover and comparative statics
295–300	§3.2	Case 4: Swedish hair salons
301	Back matter	Source appendix divider; derivations integrated in text
302–309	§1.2	Original appendix: CES entry and price leadership
310–313	§2.1	Original appendix: market-size elasticity
314	§2.2	Original appendix: Leibniz rule
315–316	§2.4	Original appendix: exact CES price index

Lecture source and cited work

Hui Shen, *Dynamic Aspects of Imperfect Competition*, slides dated October 14, 2026. The supplied annotated file contains 316 physical pages and an appendix.

The original lecture attributes the models and cases to von Stackelberg (1934), Schelling (1960), Bresnahan and Reiss (1990, 1991), Berry and Waldfogel (1999), Chamberlin (1933), Spence (1976), Dixit and Stiglitz (1977), Sutton (1991), Ellickson (2007), and Asplund and Nocke (2006). The price-leadership appendix cites Etro (2008), *Stackelberg Competition with Endogenous Entry*, *Economic Journal* 118, 1670–1697. Historical estimates and case numbers above are the lecture's reported illustrations.